Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

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Abstract

We explore how generating a chain of thought?a series of intermediate reasoning steps?significantly improves the ability of large language models to perform complex reasoning. In particular, we show how such reasoning abilities emerge naturally in sufficiently large language models via a simple method called chain-of-thought prompting, where a few chain of thought demonstrations are provided as exemplars in few-shot prompting. Experiments on arithmetic, commonsense, and symbolic reasoning benchmarks show that chain-of-thought prompting dramatically outperforms standard prompting. On the GSM8K benchmark of math word problems, chain-of-thought prompting with PaLM 540B elevates solve accuracy from 17.9% to 58.1%, setting a new state of the art.

1. Introduction

Large models excel at intuitive translation and pattern matching, but fail on multi-step reasoning tasks like math word problems. Chain of thought mimics human deliberative problem-solving by breaking tasks down into structured intermediate thoughts: problem decomposition, deduction, and calculation.

2. Benchmarks and Results

We evaluated chain-of-thought prompting across GSM8K, SVAMP, and ASDiv arithmetic benchmarks, as well as CSQA and StrategyQA commonsense reasoning. On GSM8K, 8-shot chain-of-thought prompting achieved 58.1% accuracy with PaLM 540B, compared to 17.9% for standard 8-shot prompting. Chain-of-thought prompting exhibits an emergent ability: models with fewer than 100 billion parameters show zero or negligible benefit from chain-of-thought, whereas models larger than 100B parameters demonstrate dramatic leaps in performance.

3. Limitations

A major limitation is that chain of thought can produce plausible-sounding but factually unfaithful rationales where the model derives a correct answer through invalid mathematical reasoning.